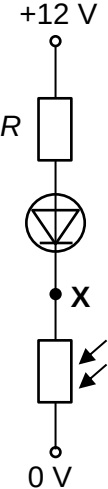
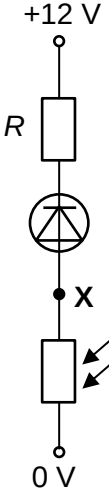
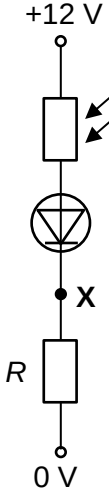
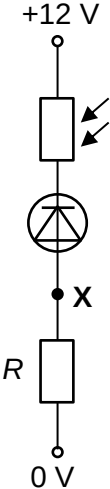


22	In bright light, a light-dependent resistor (LDR) has a resistance of R. It is connected in series with an ideal diode and a fixed resistor of resistance R. An ideal diode has zero resistance in the forward direction and infinite resistance in the reverse direction. In which arrangement will the potential at X increase when the circuit is moved to a darker environment?			
	A	B	C	D
	 Diagram A shows a series circuit. At the top is a +12 V terminal. Below it is a fixed resistor labeled R. Below the resistor is an ideal diode symbol pointing downwards. Below the diode is a point labeled X. Below X is an LDR symbol (rectangle with two arrows pointing towards it). At the bottom is a 0 V terminal.	 Diagram B shows a series circuit. At the top is a +12 V terminal. Below it is a fixed resistor labeled R. Below the resistor is an ideal diode symbol pointing upwards. Below the diode is a point labeled X. Below X is an LDR symbol (rectangle with two arrows pointing towards it). At the bottom is a 0 V terminal.	 Diagram C shows a series circuit. At the top is a +12 V terminal. Below it is an LDR symbol (rectangle with two arrows pointing towards it). Below the LDR is an ideal diode symbol pointing downwards. Below the diode is a point labeled X. Below X is a fixed resistor labeled R. At the bottom is a 0 V terminal.	 Diagram D shows a series circuit. At the top is a +12 V terminal. Below it is an LDR symbol (rectangle with two arrows pointing towards it). Below the LDR is an ideal diode symbol pointing upwards. Below the diode is a point labeled X. Below X is a fixed resistor labeled R. At the bottom is a 0 V terminal.

L2 Answer: A

Diodes in **options B and D** are in **reverse biased** connection (like an open circuit where the diode is).

- No current flows → zero p.d. across the resistance → potential at X = 0 V in both bright and dark conditions, i.e. no change in potential at X for options B and D.
- Eliminate options B and D.

Diodes in **options A and C** are in **forward biased** connection (like zero resistance where the diode is).

- Current flows → non-zero p.d. across the resistance.

Since **LDR's resistance increases** when moved into the dark, by Potential Divider Principle, the p.d. across the LDR will increase.

Hence in option A potential at X will increase, while in option C potential at X will decrease.